

TUGAS 2

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SISTEM INFORMASI

Tugas 2

1. $x_1 + x_2 + x_3 = 4$
 $x_3 = 2$
 $(a^2 - 4) \cdot x_3 = a - 2$

c) Substitusi $x_3 = 2$ ke dalam persamaan 3 :
 $(a^2 - 4) \cdot 2 = a - 2$
 $2a^2 - 8 = a - 2$
 $2a^2 - a - 6 = 0 \rightarrow \text{Faktorkan } a = 2 ; a = -\frac{3}{2}$

e) Substitusi $a = 2$ ke persamaan 3 :
 $(2^2 - 4) \cdot 2 = 2 - 2$
 $0 \cdot 2 = 0$
 $0 = 0$

e) Substitusi $a = -\frac{3}{2}$ ke persamaan 3 :
 $(\left(-\frac{3}{2}\right)^2 - 4) \cdot 2 = -\frac{3}{2} - 2$
 $\frac{-7}{4} \cdot 2 = -\frac{7}{2}$
 $\frac{-7}{2} = -\frac{7}{2}$

Maka dapat disimpulkan sistem memiliki solusi tak hingga banyak.

$$2. \quad x + 2y - 3z = 0$$

$$3x - y + 5z = 0$$

$$4x + y + (k^2 - 14)z = 0$$

$$A = \left(\begin{array}{ccc|cc} 1 & 2 & -3 & 1 & 2 \\ 3 & -1 & 5 & 3 & 1 \\ 4 & 1 & k^2 - 14 & 4 & 1 \end{array} \right)$$

$$\det(A) = (1 \cdot -1 \cdot k^2 - 14) + (2 \cdot 5 \cdot 4) + (-3 \cdot 3 \cdot 1) - (-3 \cdot -1 \cdot 4) + (1 \cdot 5 \cdot 1) + (2 \cdot 3 \cdot k^2 - 14)$$

$$\det(A) = -k^2 + 14 + 40 - 9 - (12 + 5 + 6k^2 - 84) = -7k^2 + 122$$

Karena solusi tak hingga banyak :

$$\det(A) = 0$$

$$-7k^2 + 122 = 0$$

$$k^2 = 16$$

$$k = \pm 4 \rightarrow k = 4 ; k = -4$$

3.

$$B = \begin{pmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \quad \det(B) \dots ?$$

$$\det(B) = \begin{vmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{vmatrix}$$

$$c) R_2 \leftarrow R_2 + \frac{2}{3} R_1 = \begin{bmatrix} 3 & -2 & 0 \\ 0 & \frac{5}{3} & 0 \\ 0 & 0 & 5 \end{bmatrix} \rightarrow \text{Matrix Segitiga Atas.}$$

$$d) \det(B) = 3 \cdot \frac{5}{3} \cdot 5$$

$$\det(B) = 75/3$$

$$\det(B) = 25$$

4.

$$A = \begin{bmatrix} a & 0 & b & | & 2 \\ a & a & 4 & | & 4 \\ 0 & a & 2 & | & b \end{bmatrix} \rightarrow A = \begin{bmatrix} a & 0 & b & | & a & 0 \\ a & a & 4 & | & a & a \\ 0 & a & 2 & | & 0 & a \end{bmatrix}$$

$$\det(A) = (a \cdot a \cdot 2) + (0 \cdot 4 \cdot 0) + (b \cdot a \cdot a) - (b \cdot a \cdot 0) + (a \cdot 4 \cdot a) + (0 \cdot a \cdot 2)$$

$$\det(A) = 2a^2 + 0 + a^2b - 0 + 4a^2 + 0$$

$$\det(A) = 2a^2 + a^2b - 4a^2$$

$$\det(A) = a^2(b-2)$$

Agar solusi tunggal maka $\det(A) \neq 0$
 $a^2(b-2) \neq 0$

$$i) a \neq 0$$

$$ii) b \neq 2$$

$$5. \quad A = \begin{vmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{vmatrix} \quad \Rightarrow \quad \begin{array}{ll} M_{11} = 1 & a_{11} = 4 \\ M_{12} = -2 & a_{12} = 0 \\ M_{13} = 2 & a_{13} = 1 \end{array}$$

$$\det(A) = a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + a_{13} \cdot C_{13}$$

$$\det(A) = 4 \cdot (-1)^2 \cdot 1 + 0 \cdot (-1)^3 \cdot -2 + 1 \cdot (-1)^4 \cdot 2$$

$$\det(A) = 4 + 0 + 2$$

$$\det(A) = 6.$$